Fiscal Year:	FY 2011	**Task Last Updated:**	FY 04/14/2011
PI Name:	Paul, Anna-Lisa Ph.D.		
Project Title:	The biological impact of spaceflight and extraterrestrial environments on molecular signaling and gene expression in plants		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	06/09/2009	**End Date:**	06/08/2012
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Agata Zupanska - Postdoctoral Research Associate ; Justin Goodwyn - Undergraduate Research Student ; Fiona Denison - Postdoctoral Research Associate ; Lauren Frizzle - Undergraduate Research Student ; Claire Amalfitano - Biological Scientist		
COI Name (Institution):	Ferl, Robert (University of Florida)		
Grant/Contract No.:	NNX09AL96G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overall goal of our program and this proposal is to understand the fundamental molecular responses of plants to spaceflight environments, with particular emphasis on the changes in gene expression and signal transduction that accompany spaceflight experience. We have used genetically engineered plants and molecular tools to monitor plant responses to spaceflight and space-related analog environments and have developed specific hypotheses regarding plant signaling responses that are affected by spaceflight. The focus of this proposal is to build on these extensive insights to develop the next generation of biological assays and hardware concepts to address these hypotheses in ground and analog studies. In the process, these new biological and analytical tools will be integrated into functional concepts than can be readied for future deployment in a range of spaceflight-related opportunities. The biology will include new biosensors to test hypotheses regarding the mechanisms by which molecular signals are conveyed in response to spaceflight and analog environments. Some biosensors will be linked to Green Fluorescent Protein variants to facilitate multiplexed telemetric data collection. The analytical techniques will make use of current spaceflight imaging technologies and explore advanced imaging concepts. This combination of biosensors and imaging will enable robust robotic experiments with flexible future deployment options, enabling experiments that address science needs to survey molecular responses in a variety space venues and situations. The expected significance of this project is that it will advance the fundamental understanding of spaceflight molecular signaling in spaceflight-relevant analogs and then provide the means to transition these studies for tests in bona fide spaceflight environments as opportunities develop. Thus the objectives of this proposal are completely aligned with the request of the NRA to investigate the nature of cellular receptors, signal transduction pathways and changes in gene expression in response to spaceflight-relevant environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The expected significance of this project is that it will advance the fundamental understanding of molecular signaling in spaceflight-relevant analogs and then provide the means to transition these studies for tests in bona fide spaceflight environments as opportunities develop. Thus the objectives of this proposal are completely aligned with the request of the NRA to investigate the nature of cellular receptors, signal transduction pathways and changes in gene expression in response to spaceflight-relevant environments.
Task Progress:	The second year of the grant was focused on two fronts: first, several Launch opportunities to the ISS associated with the PIs' other NASA grant enabled us to test some of the biological tools that were calibrated in ground studies in the first year, and described in last year's report. Second, Arabidopsis mutants were used to assay the role of certain genes associated calcium and phosphorylation mediated signal transduction in stress response. In addition, the development of additional biosensors and signaling mutants continues. Some details include: Extensive optimization of RNA analyses for all research in the laboratory, including utilizing BioAnalyzer technology for evaluations of RNA integrity. Seed lines - In 2011, several of the lines characterized in 2010 were included in the Flight opportunity: pTCS::GFP, ARR-family mutant, AXR3 and EIR1. Those data are currently being analyzed. Verification of calcium signaling mutations - the verification of mutants is to confirm that the Arabidopsis lines carry the T-DNA insertion, and that it is positioned to abolish gene function. The presence of the pBIN-pROK2 insert was verified using a set of primers derived from the sequence found on the Salk website (http://signal.salk.edu/). The location of the insert was verified with a primer derived from the genomic DNA of Arabidopsis upstream of the left insert flanking region (sequence provided by TAIR- arabidopsis.org) and LBa1, provided by the Salk site. DNA was isolated form mutant and wild type plants and used as templates for amplification with primers to regions within the coding region of the target gene and to sequences unique to the T-DNA insertion. Amplified sequence form the mutant plants (CBL1, CBL9, CIPK3, CIPK23) contain discrete amplified sequences corresponding to the T-DNA insert, whereas the wild-type (WT) does not. Amplification products confirm that the genes encoding the calcium signaling genes of interest in the mutant plants is disrupted by the T-DNA insertion. Production of Biosensor plants in mutant backgrounds - Once verified, the calcium signaling mutants are used as recipients for pollen from plants homozygous for several reporter gene constructs. Cross pollination is carried out by selecting flowers whose sepals have not yet opened, at flower stage 10. The stamens and part of the petals and sepals are removed to prevent self-pollination. At flower stage 11, they are pollinated with stamen from one of the GFP reporter-gene donors, then covered with a small piece of plastic wrap from prevent the pistil from drying or being contaminated with any other pollen source. Currently we have completed six of the 12 targeted lines. Phosphorylation mediated signal transduction - The involvement of phosphorylation events in signal transduction also appear to be strongly involved is a variety of stress responses pathways, and we are exploring phosphorylation-mediated events in addition to the calcium mediated events. The initial experiments with T-DNA insertion mutants examined the impact of the mutation on the expression of a number of genes associated with abiotic stress that involves a phosphorylation-mediated step in its pathway. The majority of these experiments were conducted by Dr. Fiona Denison, Dr. Agata Zupanska, and Mr. Justin Goodwyn (undergraduate). Thus far, seven different target genes and eight different phosphorylation mutants have been evaluated. Parabolic Flight Data Analyses - In addition to the conducting of new experiments salient to the current grant, the grant also supported new data analyses of parabolic flight data collected under a previous award. Briefly, in a succession of experiments on NASA's KC-135 and C-9 parabolic aircraft, Arabidopsis plants were exposed to a series of flight sets that provided controlled, replicated exposure to parabolic flight. Gene expression profiling of these plants reveals that parabolic flight, each presenting 40 durations of alternating microgravity and hypergravity, can induce wide-scale changes in gene expression patterns that stand the statistical tests of replication on three different flight days. The earliest response is characterized, after 20 parabolas, by a prominence of genes primarily associated with signal transduction. After 40 parabolas this prominence is largely replaced by a variety of genes associated with biotic and abiotic stimuli and stress. Among these responses, three metabolic processes stand out in particular: the induction of auxin metabolism and signaling, many of which are known to be involved in gravity sensing in plants; the differential expression of genes associated with calcium mediated signaling, and the repression of genes associated with disease resistance and cell wall biochemistry. Changes in auxin-related gene expression were also recorded by reporter genes tuned to auxin signal

	pathways. These data demonstrate that the parabolic flight environment is appropriate for molecular biology research involving the transition to microgravity, in that with replication, proper controls and analyses, gene expression changes can be observed in the timeframes of typical parabolic flight experiments. These analyses facilitated the publication of those data in this period, and the manuscript is currently under review in Astrobiology.
Bibliography Type:	Description: (Last Updated: 11/20/2025)
Abstracts for Journals and Proceedings	Paul A-L, Zupanska A, Osterow DT, Zhang Y, Shanker S, Farmerie WG, Ferl RJ. "The impact of Spaceflight on Arabidopsis: Deep Sequencing and DNA Arrays as Collaborative Readouts of the Transcriptome of Arabidopsis Seedlings and Undifferentiated Cells in Space." Presented at 26th annual meeting of the American Society of Gravitational and Space Biology, National Harbor, MD, November 4-7, 2010. Program and abstracts. 26th annual meeting of the American Society of Gravitational and Space Biology, p. 24. , Nov-2010
Abstracts for Journals and Proceedings	Paul A-L. "Analog settings and biosensor technology: tools for exploring the biological impact of extraterrestrial environments." Invited seminar. Tryon Seminar Series, Department of Cell Biology, Microbiology, and Molecular Biology, University of South Florida, Tampa, FL, September 2010. Department of Cell Biology, Microbiology, and Molecular Biology, University of South Florida, Tryon Seminar Series , Sep-2010
Abstracts for Journals and Proceedings	Paul A-L, Ferl RJ. "Exploring space between parabolic flight and orbital environments - altered gene expression in Arabidopsis thaliana." Oral presentation. 2011 Next-Generation Suborbital Researchers Conference, University of Central Florida, Orlando, Florida, February 28-March 2, 2011. 2011 Next-Generation Suborbital Researchers Conference, University of Central Florida, Orlando, Florida, February 28-March 2, 2011. http://www.swri.org/9what/events/confer/nsrc/2011/abstracts/crew/28845798-2214460.pdf , Mar-2011
Articles in Peer-reviewed Journals	Visscher AM, Paul A-L, Kirst M, Guy CL, Schuerger AC, Ferl RJ. "Growth performance and root transcriptome remodeling of Arabidopsis in response to Mars-like levels of magnesium sulfate." PLoS One. 2010 Aug 23;5(8). pii: e12348. PMID: 20808807 , Aug-2010
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